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RESEARCH

Enhancing IoT network security with self-supervised multi-residual attention graph networks

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Abstract

An Intrusion Detection System is crucial for securing Internet of Things. The conventional Intrusion Detection System technologies face challenges in detecting minority-class intrusions due to class imbalance and limited ability to model intricate dependencies in network traffic. To address these challenges, this research provides an intelligent framework to detect network intrusion in an Internet of Things network named Self-Supervised Multi-Residual Attention Graph Network (SS-MRAGNet). Specifically, SS-MRAGNet integrates node and edge aggregation with a multi-residual attention mechanism to learn complex structural dependencies within network traffic data. The framework incorporates self-supervised pre-training with pseudo-labeling and edge augmentation to address the class imbalance challenges and detect minority-class intrusions accurately. A minimal Random Under Sampling strategy is applied only within training folds to stabilize optimization, without serving as the primary imbalance-handling mechanism. Then, the model performs iterative self-training to generate high-confidence predictions, which are then used to pseudo-label unlabeled data. After the iterative process, predictions are refined. Rademacher complexity framework and convergence proofs are utilized for theoretical analysis to analyze its performance in noisy and imbalanced data. On validation, CIC-Darknet2020, CSE-CIC-IDS2018, UNSW_NB15, and Ton-IoT datasets result in 98.25% accuracy, 98.21%, precision, 98% recall, and 98.1% F1-score, outperforming state-of-the-art models. Complexity analysis further demonstrates its suitability for near real-time deployment at the fog/gateway level. This study proposes a novel graph-based IDS framework and provides practical insights into handling class imbalance, highlighting its potential for secure and scalable IoT intrusion detection.

Keywords Class Imbalance · Edge Augmentation · Graph Neural Network · Network Intrusion Detection System · Node and Edge Aggregation · Self-Supervised Learning · Pseudo-Labeling

1 Introduction

Due to the rapid evolution of technology, the Internet of Things (IoT) has expanded significantly and now supports a wide range of applications across healthcare, smart cities, industrial automation, transportation, and critical infrastructure. IoT environments involve interconnected devices that continuously collect, exchange, analyze, and transmit data across distributed networks [1]. However, the heterogeneous and dynamic nature of IoT traffic also increases vulnerability to cyberattacks, unauthorized access, data breaches, and malicious intrusions. Therefore, ensuring robust security mechanisms has become a critical requirement for maintaining reliable and trustworthy IoT infrastructures [2].

Intrusion means any unauthorized entry or malicious activity that is designed to harm network integrity, alter network operations, steal information or disrupt the service [3]. The nature of intrusions is very dynamic and constantly changing, making it difficult for conventional security systems to accurately detect advanced and infrequent attacks. Hence, IoT traffic monitoring, abnormal behavior detection, and cyber threats protection of network resources have become one of the most critical security mechanisms, known as Intrusion Detection Systems (IDSs) [4].

Although conventional IDS frameworks have demonstrated effectiveness in identifying common attacks, they face substantial limitations when applied to real-world IoT environments [5]. One of the major challenges is the

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severe class imbalance present in modern intrusion datasets, where malicious samples-particularly minority-class attacks such as botnets, infiltration attacks, or zero-day intrusions-appear significantly less frequently than benign traffic. This imbalance causes conventional machine learning and deep learning models to become biased toward majority classes, thereby reducing the detection capability for rare but critical attacks [6].

The traditional imbalance-handling methods primarily include oversampling, undersampling or cost-sensitive learning strategies [7]. But too much oversampling can cause redundant or synthetic patterns in the data, which can cause overfitting, while too much undersampling can cause informative majority class samples to be removed and can change the original data distribution [8]. Moreover, some of the existing IDS models are not able to capture complex relationships between IoT devices in terms of their structural dependency and communication pattern and therefore they are unable to be generalized in case of dynamic and heterogeneous network conditions [9].

Recent graph-based intrusion detection approaches have attempted to model IoT traffic as graph structures, where nodes represent devices and edges represent communication interactions. Graph Neural Networks (GNNs) have shown promising capability in learning topological relationships and contextual dependencies within such environments. Nevertheless, many existing graph-based IDS methods focus primarily on node-level representations while overlooking critical edge-level information and minority-class structural relationships. Additionally, most existing methods still heavily depend on large labeled datasets and conventional resampling mechanisms, limiting their effectiveness in practical deployment scenarios [10].

To overcome these limitations, this research proposes a novel framework named Self-Supervised Multi-Residual Attention Graph Network (SS-MRAGNet) for intelligent intrusion detection in IoT networks. The proposed framework integrates node-edge aggregation with a multi-residual attention mechanism to effectively capture both local and global structural dependencies within IoT traffic graphs. Unlike conventional IDS approaches that primarily rely on aggressive resampling strategies, SS-MRAGNet addresses class imbalance mainly through self-supervised representation learning mechanisms, including pseudo-labeling, edge augmentation, and iterative self-training.

Specifically, the proposed framework strengthens minority-class representation by augmenting graph connectivity among sparsely connected minority nodes and learning discriminative representations from both labeled and unlabeled traffic data. A lightweight Random Under Sampling (RUS) procedure is utilized only within fold-specific training subsets to stabilize optimization during cross-validation and does not constitute the primary imbalance-handling mechanism of the framework. This distinction enables the proposed model to preserve original data characteristics while improving robustness against imbalance and noisy IoT traffic conditions.

Moreover, the integration of self-supervised pre-training enables SS-MRAGNet to learn meaningful feature representations without relying extensively on labeled datasets. The iterative pseudo-label refinement mechanism further improves minority-class detection capability by progressively enhancing the quality of learned representations during training. In addition, the multi-residual attention mechanism selectively emphasizes informative node-edge interactions while mitigating information loss during deep graph propagation.

The effectiveness of the proposed framework is validated using multiple benchmark IoT intrusion datasets, including CIC-Darknet2020, CSE-CIC-IDS2018, UNSW-NB15, and ToN-IoT datasets. Experimental outcomes demonstrate that SS-MRAGNet achieves superior performance in terms of accuracy, precision, recall, F1-score, robustness, and computational efficiency compared with several state-of-the-art IDS models. The framework also demonstrates strong resilience under noisy and highly imbalanced IoT environments, highlighting its suitability for scalable and near real-time intrusion detection deployment.

The major contributions of this research are summarized as follows:

- Development of a graph-based IDS framework integrating node-edge aggregation with a multi-residual attention mechanism for enhanced IoT intrusion detection.
- Introduction of self-supervised representation learning strategies, including pseudo-labeling and edge augmentation, to improve minority-class intrusion detection without relying primarily on aggressive resampling methods.
- Enhancement of minority-class structural representation through graph connectivity augmentation and iterative self-training refinement.
- Validation of robustness and generalization capability under noisy and highly imbalanced IoT conditions using multiple benchmark datasets.
- Demonstration of computational feasibility for practical edge/fog-level IoT security deployment through complexity and statistical analyses.

In the remaining sections, Section 2 provides the analysis of the literature survey related to existing IDS. Section 3 provides the theoretical analysis of the SS-MRAGNet**.** The proposed methodology of the SS-MRAGNet framework is presented in Section 4. The experimental results are

provided in Section 5.1, and the discussion with the future scope is provided in Section 6.1. Finally, the conclusion is presented in Section 9..

2 Literature survey

The literature survey on various IDS that have existed recently and the research gaps are discussed in this section.

2.1 Related works

Sanju [11] established a Modified Metaheuristics with Weighted Majority Voting Ensemble Deep Learning (MM-WMVEDL) model to identify the intrusion in the environment of IoT. The two models like Gated Recurrent Unit (GRU) model and Long Short-Term Memory (LSTM) were used as recurrent neural networks (RNNs), employed to detect different kinds of attacks. Wavelet-dependent feature extraction was incorporated into the model to improve the extracted features' discriminative power. Min–max normalization was involved for normalizing the input. Harris hawk optimization-based elite fractional derivative mutation (HHO-EFDM) technique was utilized for feature selection. One advantage of the method is its handling ability of multi-modal network traffic data. Another advantage is that it utilizes global and local patterns by utilizing wavelet-dependent feature extraction. A DL based Intrusion Detection System (DL-based IDS) was established by Bakhsh et al. [12] through the use of Random Neural Networks (RandNN), LSTM, and Feed Forward Neural Networks (FFNN) for securing networks of IoT against cyberattacks. The complex traffic patterns in the network of IoT were handled through FFNN, while long-term traffic dependencies were captured through LSTM. The RandNN model utilized its ability of data learning to adapt and learn from the data of the network. Cybersecurity was improved through these algorithms by protecting the sensitive data and enabling mechanisms of defense against cyber threats. The method enhanced entire performance of IDS by understanding the complicated interaction between output categorizations and characteristics of input. The hyperparameters utilized for the model were a 0.1 dropout rate, a 0.001 weight loss, 128 batch sizes, and 100 epochs.

Zhang and Liu [13] implemented the Improved Conditional Variational Autoencoder (ICVAE) and Borderline Synthetic Minority Oversampling Technique (BSM), named ICVAE-BSM, which was based on IoT for intrusion detection. The Conditional Variational Autoencoder (CVAE) was implemented with an auxiliary network for adjusting the output of the encoder's probability distribution. For BSM, a dimensionality reduction algorithm was utilized to oversample boundary in the space of latent variables. The same class samples were clustered, and the different class samples were scattered. After that, the latent variables at the edge were synthesized and fed into ICVAE's decoder to produce new samples for balancing the dataset. Through the Softmax classifier, the real samples were mixed with new samples to improve the encoder's ability for generalization. At a ratio of 4:1, the dataset was divided. For testing the model, 10% of the dataset was utilized. The approach was also used to identify minority intrusions from IoT unbalanced samples. An improved dynamic sticky binary particle swarm optimization (IDSB-PSO) is designed by Sarwar et al. [14] to select the features for an enhanced system of anomaly detection. A reduction strategy of the dynamic search space and dynamic parameters was implemented to improve the searching capability of sticky binary particle swarm optimization (SBPSO). Iteratively, a mask was applied to the features after some generations. It reduced the search space to detect better solutions within the space of smaller search space. The evaluation parameters were the acceleration constant increased from 0.5 to 2.5 and then decreased from 2.5 to 0.5, the mask updating parameter was 0.25, and the swarm size was 20. A deep learning (DL) model in IoT systems for intrusion detection was implemented by Elnakib et al. [15], named an enhanced anomaly-based Intrusion Detection Deep learning Multi-class classification model (EIDM). On the cloud server, the five deep learning models were trained and employed in a gateway node. The network traffic classification occurred in the gateway node. The evaluation parameters were the number of epochs 100, the kernel size 20, and the learning rate 0.0005.

Alghanam et al. [16] developed a pigeon-inspired optimization (PIO), an enhanced version with a local search algorithm called LS-PIO for intrusion detection in IoT networks. A multiple one-class classifier-based ensemble learning was utilized to enhance the performance of this method. The ensemble IDS model integrated Local Outlier Factor (LOF), one-class support vector machine (OC-SVM), and Isolation Forest (IF), which was utilized after the phase of feature selection. The evaluation parameters were a number of iterations 400, the population size 128, and the number of runs 20. A machine learning-based intrusion detection model was illustrated by Roy et al. [17] for the identification of anomalies and cyber-attacks in the networks of IoT. For detecting features, techniques like dimensionality reduction, removal of multicollinearity, and sampling was deployed for intrusion detection through minimum data and training time. Principal Component Analysis (PCA) was utilized for reducing dimensionality. The parameter that was set for estimation was the number of principal elements was set to 10.

A Multi-Objective PSO-based Intrusion Detection System (IMOPSO-IDS) was implemented by Subramani and Selvi [18] for identifying intruders based on IoT in sensor networks. An algorithm for feature selection, which was based on rules and Multi-Objective PSO, and an improved Multiclass Support Vector Machines classification algorithm for classification, which was based on a rule-based classification strategy, were implemented for intelligent IDS. Using a particle swarm-dependent method, the multi-objective dependent issues were solved. The simulation evaluation parameters were the number of sensor nodes 100–1000, packet size 4098 bits, and network area 200 m × 200 m. Wang et al. [19] designed a knowledge-distillation-based IoT intrusion detection model called BT-TPF to detect attacks in the network. For the reduction of feature dimensionality, a Siamese network was leveraged, and a large-scale Vision Transformer was also employed during training to direct a small-scale Poolformer model. An improved technique of BERT-of-Theseus was implemented to reduce the gap that exists between the outcome of the predecessor and the successor models. The small final model reduced the parameter quantity through knowledge distillation and maintained high accuracy on detection. The evaluation parameters were the convolutional layer's size of kernel was 2, the convolution step in patch embedding was 2, and the number of hidden layer neurons was 5.

Basati and Faghih [20] developed a deep feature extraction (DFE) approach, an accurate Network intrusion detection system (NIDS) that utilizes efficient and lightweight neural networks depending on the feature extraction idea. The network's input vector was permuted in a 3-dimensional space, and brought its individual values closer together. The discriminative features were extracted through a smaller quantity of layers, and for this, large 3D or 2D convolution filters were not utilized. The evaluation parameters were batch size 32, kernel size 2, and learning rate 0.01. Malibari et al. [21] established the MDLIDS-SSE technique, an intrusion detection system enabled by DL to secure a smart environment. To choose an optimal feature subset, an improved arithmetic optimization algorithm based feature selection (IAOA-FS) approach was enabled and for data pre-processing, Z-score normalization was deployed. The implementation of deep wavelet neural network named DWNN and an optimization technique called quantum-behaved particle swarm optimization named QPSO detected and classified the intrusions. Through QPSO, the tuning parameters were performed. The parameters for evaluation were epoch count 50, batch size 64, and learning rate 0.01.

A NSBPSO-DCNN, a deep convolutional neural network (DCNN) based neighborhood search-based particle swarm optimization algorithm was implemented by Baniasadi et al. [22] for detection of intrusion through increasing the PSO algorithm's exploration/exploitation. To attain better accuracy, NSBPSO was used to optimize the parameters of the deep architecture. In the DCNN, the fully connected multilayer perceptron (MLP) was used, where biases and optimized weights using NSBPSO. The settings of parameters for evaluation were number of iterations 300, population size 100 and inertial movement rate 0.08. An approach for autoencoder-based identification is established by Khan et al. [23] through leveraging recurrent and convolutional networks to estimate cyber threats in IIoT. The representations of data features' latent were studied through implementing a two-step sliding window (SW). Points of malicious were varied from the raw time into a fixed series of length utilizing the SW initial-step. For attack events categorization, the extracted features of spatial and temporal were leveraged by networks that are in fully connected. The settings of parameters for evaluation were optimizer Adam, learning rate 0.001, and epochs 100.

A federated-simple recurrent units (Federated-SRUs), an intrusion detection system (IDS) model was implemented by Khan et al. [24] for securing IoT-dependent industrial control systems (ICSs). To alleviate the issue of gradient vanishing and minimize computational cost, an enhanced simple recurrent unit framework was utilized by the federated-SRUs IDS model. In the architecture of federated learning (FL), data aggregation was performed by using multiple communication rounds. This architecture permitted the stakeholders to construct an extensive IDS model. The parameter settings of the approach were learning rate 0.001, number of malicious clients 8, and local epoch 5. Bian and Liu [25] developed a gaussian mixture cramer-wold auto-encoder (GMCWAE) to learn the discriminative and interpretable network traffic's low-dimensional representations to enhance the classifiers' detection performance. Light Gradient Boosting Machine (LightGBM) was integrated to identify intrusive traffic. GMCWAE learned low-dimensional representations across all the utilized datasets. The parameter settings of the approach were optimizer Adam, layer reduction rate 2, and activation function linear.

A federated-contrastive-learning approach (FeCo) was established by Wang et al. [26] for intrusion detection by utilizing federated learning to reduce the privacy concerns of the users. Based on contrastive learning, a representation learning method was used to develop an accurate framework for the benign class. To minimize computation time and neglect overfitting, a scheme of two-step selection scheme was implemented. FeCo tried to decrease the distance between the samples of malicious and benign. The parameter settings of the approach were learning rate 0.001, number of clients 50, and size of input layer 112. Mwaura et al. [27] implemented a Transfer learning TinyML-based IDS model along with feature selection to detect IoT device

attacks like zero-day attacks and malware attacks. The DDoS attack identification model was utilized to design this approach. Using two different groups of datasets, the model was evaluated, and it was changed into a format that was compatible with a microcontroller. The parameter settings of the approach were batch size 256, learning rate decay 0.1, and learning impulses 0.9.

Jianping et al. [28] developed a Federated Graph Attention Network (FedGAT), an attention-based GNN model to identify network attacks that were cross-department and cross-level. It protected the privacy of data over a distributed service to enable training of collaborative training. In chronological order, the network traffic details were organized and depending on log density, a graph structure was constructed. FedGAT model was utilized to determine the nodes' interaction, thus increasing the interactions of the internal network's precision. The method achieved efficient results when compared with other baseline detection methods. Sun et al. [29] illustrated a hierarchical attention-based dynamic GNN algorithm (HADGA) to detect a botnet in the network of dynamic IoT. Through the dual-attention model, this approach decoupled spatiotemporal attention. Inductively, the nodes' spatial embeddings were generated by fusing GraphSAGE and the neighbor attention module. The temporal evolution details are captured by the temporal attention module through weighting the nodes' historical representations flexibly. On evaluation, the algorithm attained accuracy of 97.9%, which is better than prior approaches.

A 2-layer control area network graph attention networks (CAN-GAT-2) model was established by Xiao et al. [30] to detect anomalies in the network of vehicles. It also presented a graph attention, graph convolution based detection model. Using the CAN bus traffic, a graph was constructed to seize the variation in the traffic bytes and enhance the efficiency and detection accuracy. The quantitative analysis demonstrated that the highest performance improvement was achieved by the CAN-GAT-2. Wang et al. [31] implemented GADMCLG, a graph anomaly detection based multiscale contrastive self-supervised learning approach that incorporated global-level contrast that was in the form of subgraph-subgraph and local-level contrast that was in the form of node–node. Localized abnormal details were mined by the former and then global patterns of anomaly were captured. The h-order nearby sampling of a subgraph was adopted by the contrast of subgraph-subgraph. This provided interpretability and reduced extraneous noise for the identification of anomalies. In the scenarios of few-shot, Mao et al. [32] illustrated a label-aware federated graph contrastive learning model called FeCoGraph to detect the intrusion. The flow embeddings were processed directly using a line graph. Graph contrastive learning was formulated to implement label details, permitting embeddings as intra-class than inter-class. FL was utilized to enhance the NIDS's scalability while safeguarding the privacy of data. The efficiency was demonstrated by achieving better accuracy in categorization. The outline of related works is provided in the supplementary file (Part A).

Most of the traditional IDS models like EIDM, DFE, MDLIDS-SSE overlook the imbalance problem, depending on large datasets or implicit sampling, which leads to bias toward majority classes. Sampling-based and re-weighting methods (e.g., ICVAE-BSM, IMOPSO-IDS) increase minority recall but risk overfitting or data loss. Recent self-supervised and graph-based methods like GADMCLG, FeCoGraph, HADGA, FedGAT increasingly leverage contrastive learning, attention, or topological cues to implicitly re-weight minority nodes. However, these still depend on balanced labeled data or lack explicit graph augmentation. In contrast, SS-MRAGNet integrates pseudo-labeling, edge augmentation, and multi-residual attention, a combination that jointly leverages self-supervised representation learning and graph-structural cues to balance the minority-class representation without oversampling, thereby outperforming all baselines across multiple IoT datasets.

2.2 Research gaps

Even though there are various IDSs that have shown effective results for IoT networks, many gaps are found in the traditional techniques. Some of them are discussed below:

Inadequate intrinsic handling of class imbalance: Most existing methods leverage strategies like oversampling, undersampling, and cost-sensitive learning. These strategies introduce issues like overfitting, distortion of the original data distribution, and loss of informative samples. Existing approaches lack the ability to address the imbalance intrinsically through representation learning without depending on conventional techniques of resampling. The proposed model addresses this issue of class imbalance that exists in the datasets by utilizing self-supervised representation learning mechanisms like pseudo-labeling and edge augmentation.

Insufficient integration of self-supervised learning for IDS: Existing works rarely leverage self-supervised learning in a unified manner to utilize unlabeled data effectively for improving feature representation and generalization. The proposed model incorporates a self-supervised pre-training phase using pseudo-labeling and auxiliary tasks to learn meaningful representations from unlabeled IoT data.

Weak representation of minority-class structures in graph space: In some cases, the minority-class nodes

are often sparsely connected in graph-based IDS models, resulting in poor feature propagation and reduced detection performance for rare attacks. They lack effective graph augmentation strategies to strengthen minority-class connectivity and representation. To address this gap, the proposed model implements an edge augmentation strategy to strengthen minority-class node connectivity.

Limited modeling of joint node and edge interactions: Most of the existing graph-based approaches focus on node-level features and neglect edge-level information to encode critical communication patterns in the IoT networks. This restricts the ability of the approaches to capture complex structural dependencies needed for accurate intrusion detection. The proposed SS-MRAGNet overcomes this by modeling both node and edge features through a multi-residual attention mechanism, which enables effective aggregation of structural and relational information.

Security Vulnerabilities and Reliability in IDS Models: Existing methods exhibit unstable learning in the presence of dynamically evolving attack patterns and noisy traffic conditions. Their consistent performance is limited since there is an absence of robust feature learning and adaptive refinement strategies, posing a significant security risk in practical deployments. In order to overcome these issues, the developed model utilizes multiple complementary mechanisms like self-supervised representation learning, iterative self-training with confidence filtering, a multi-residual attention mechanism, and edge augmentation for structural stability.

3 Theoretical analysis

Theoretically, the proposed model is analyzed using the Rademacher complexity framework and convergence proofs.

3.1 Generalization capability

The proposed model's ability of generalization is the capability to perform well on unseen data, specifically for the case of intrusion identification in the networks of IoT. This capability is essential due to the noisy and imbalanced nature of IoT data. In these types of data, security challenges arise and there is rare minority-class intrusions detection system. To analyze this capability in a rigorous manner, this paper utilizes Rademacher complexity, a theoretical framework, to evaluate the model's measuring ability beyond training data generalization.

3.2 Rademacher complexity and generalization bound

The evaluation of Rademacher complexity depends on the richness of the function class that is fittable to random noise using a model [33]. The lower Rademacher complexity model is less overfitting to the data of training, denoting its higher capability of generalization to new data. Let H be the SS-MRAGNet model's hypothesis class, $h \in H$ be the hypothesis (model), then the definition of empirical Rademacher complexity of H is defined as:

$$\widehat{R}_n(H) = \frac{1}{n}\mathrm{E}_\sigma\left[\sup_{h\in H}\sum_{i=1}^{n}\sigma_i h(x_i)\right] \tag{1}$$

In the above equation, the independent Rademacher variables are denoted as $\sigma_i \in \{-1, +1\}$ are independent Rademacher variables with equal probability,E_σ denotes the expectation over the Rademacher variables, x_i indicates the ith training sample, and x_i's prediction or hypothesis is denoted as $h(x_i)$. With regards to SS-MRAGNet, the function space associated hypothesis class H is referred as the components of self-supervised learning and a mechanism of multi-attention. The hypothesis h's generalization error $L(h)$ is bounded by:

$$L(h) \leq \widehat{L}(h) + 2\widehat{R}_n(H) + O\left(\frac{1}{\sqrt{n}}\right) \tag{2}$$

In the above equation, the expected error is indicated as $L(h)$, computed empirical error on the training data is noted as $\widehat{L}(h)$, and the convergence rate with respect to the n sample size is denoted as $O\left(\frac{1}{\sqrt{n}}\right)$. If the Rademacher complexity $\widehat{R}_n(H)$ minimizes, which is indicated by the bound, then it also minimizes generalization error, leading to improved generalization on unseen data.

3.3 Impact of noisy and imbalanced data

By utilizing the framework of self-supervised learning and mechanism of multi-residual attention, the noisy and imbalanced data are managed effectively. Different weights are assigned to the edges and nodes according to their connection with the class of minority, efficiently minimizing the examples of majority-class impact that might overtop the process of learning. This kind of weighting in a selective manner aids in lowering the Rademacher complexity since the model focuses on learning informative patterns instead of fitting random noise.

Moreover, before classification task fine-tuning, the phase of self-supervised pre-training increases the model's generalization capability through learning strong feature representations. The regularization form of pre-training again enhances generalization and minimizes the efficient Rademacher complexity. The clustering of node and edge prediction are the self-supervised learning's supplementary tasks that improve the model's robustness to noisy data by offering extra, less sensitive to noise training signals.

On the whole, the theoretical Rademacher complexity analysis indicates that the SS-MRAGNet has a robust capability of generalization, specially in noisy and imbalanced information environments. The model architecture guarantees that the model is not overfitting to the majority class or to the noisy samples, providing a more reliable and accurate detection of intrusion in the networks of IoT.

3.4 Convergence proofs

The proposed model's convergence signifies the capability to reach a stable solution by the optimization algorithm, where additional updates do not significantly alter the model's parameters. This stable solution is significant for guaranteeing that the performance of the model is consistently better over various scenarios and datasets. Standard techniques are utilized in the theory of optimization, such as gradient descent-based optimization methods, to prove the SS-MRAGNet model's convergence.

3.4.1 Gradient descent and convergence

The gradient descent variant trains the SS-MRAGNet model, where iterative model parameters updating is performed depending on the gradients of loss function in terms of these parameters. At t iteration, the parameters of the model is considered as θ_t, and the loss function at θ_t is represented as $L(\theta_t)$, and its gradient is considered as $\nabla L(\theta_t)$. The below equation represents the update rule of gradient descent:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \tag{3}$$

here, η denotes the learning rate, which controls each update's step size using a positive scalar. The convergence of the sequence $\{\theta_t\}$ to a point θ^* (optimal parameter) is required, where the loss function gradient is zero, which is indicated as $\nabla L(\theta^*) = 0$. Let the loss function $L(\theta)$ be assumed as Lipschitz continuous and convex with a L constant, which signifies as:

$$\|\nabla L(\theta) - \nabla L(\theta^*)\| \leq \mathrm{L}\|\theta - \theta^*\| \quad \forall \theta, \theta^* \in R^d \tag{4}$$

Depending on these considerations, when the η is selected appropriately, the gradient descent algorithm's convergences to a global minimum. Particularly, For convergence, the learning rate must satisfy:$0 < \eta < \frac{2}{\mathrm{L}}$, then the $\{L(\theta_t)\}$ sequence decreases monotonically and showed convergence to the low loss function.

$$L(\theta_{t+1}) \leq L(\theta_t) - \frac{\eta}{2}\|\nabla L(\theta_t)\|^2 \tag{5}$$

This will indicate that there is a decrease in the loss function during non-zero gradient $\nabla L(\theta_t)$, resulting θ_t and θ^* optimal solution get closer.

3.4.2 SS-mragnet convergence

The extension of convergence proof in the proposed model extends to the structure of self-supervised learning and the mechanism of multi-residual attention. Every model's component contributes to the entire loss function, and the working of optimization guarantees that all self-supervised activities and attention mechanism-based parameters converge to optimal values.

In the loss function, the element of self-supervised learning develops extra terms which are Lipschitz continuous and convex, guaranteeing that the complete loss functions remains compatible with gradient descent. The loss function includes the attention mechanism related regularization terms, guaranteeing that there exists a convergent and stable optimization process.

Thus, the SS-MRAGNet's convergence is ensured through the benchmark Lipschitz and assumptions of convexity continuity. The optimization algorithm of the model will attain a stable and consistent solution, resulting in trustworthy performance against various scenarios and databases.

4 Proposed methodology

The proposed SS-MRAGNet was utilized to overcome the existing IoT intrusion detection systems" limitations that depend on labeled data and shallow feature learning. Inherently, IoT traffic is structured as a graph and thus, a GNN backbone effectively captures dependencies at node- and edge-level among devices. Selectively, the multi-residual attention mechanism emphasizes critical communication patterns while maintaining the flow of information across layers, safeguarding against over-smoothing. A self-supervised pre-training phase stimulates learning from large volumes of unlabeled IoT data, enhancing generalization and minority-attack recognition. To address class imbalance, the proposed SS-MRAGNet primarily depends on

representation learning strategies like pseudo-labeling and edge augmentation, which enhance the connectivity and feature propagation of minority-class nodes within the graph. Although a minimal RUS step is implemented during training to stabilize optimization within cross-validation folds, it is not the primary mechanism for imbalance handling. These design choices collectively make SS-MRAGNet essential for scalable and accurate IoT intrusion detection by prioritizing intrinsic learning of data structure and minority-class patterns over conventional resampling-based solutions. The proposed model's overall workflow is illustrated in Fig. 1.

4.1 Node and edge features aggregation

Within the graph IoT network, the framework of proposed begins with the aggregation of features from both edges and nodes [34]. The device interconnection and intricate contacts are captured by this aggregation. Consider G be the graph for the IoT network and it is represented as $G = (V, E)$, the node set (devices) is represented as $V = (v_1, v_2, \ldots, v_n)$ and the edges set (devices connection) is represented as $E \subseteq V \times V$. Every edge $e_{ij} \in E$ and node v_i is correspondingly related to $e_{ij} \in R^f$ and $h_i \in R^d$ feature vector.

4.1.1 Node feature aggregation

The SS-MRAGNet's aggregation of node features is formatted for capturing every node's neighborhood information in the graph. For a provided graph G, every $v_i \in V$ node is associated with the x_i feature vector. Let us consider that $h_i^{(l)}$ be the node feature at layer is updated by means of the features' aggregation of nearby edges and node through the mechanism of attention:

$$h_i^{(l)} = \sigma\left(W^{(l)} h_i^{(l-1)} + \sum_{j \in N(i)} \alpha_{ij}^{(l)} W_e^{(l)} e_{ij}\right) \tag{6}$$

where, the node 's feature vector from the previous layer is denoted as $h_i^{(l-1)}$, the beginning node feature is represented as $h_i^{(0)}$ which is equal to h_i, node's neighbors set is represented as $N(i)$, the learnable kind of weight matrices are denoted as $W_e^{(l)}$ and $W^{(l)}$, the ReLU non-linear activation function is σ and the edge e_{ij}'s attention coefficient is given as $\alpha_{ij}^{(l)}$.

4.1.2 Edge feature aggregation

The SS-MRAGNet also performs aggregation of edge features to capture the interactions among the interlinked nodes. The edge features are included to provide potentially valuable intrusion information through the essence of device interactions. Through the attention mechanism $\alpha_{ij}^{(l)}$, the aggregation of features in edge is deployed into the aggregation process of node. The model performs weight assignment to the classification task relevance-based edges.

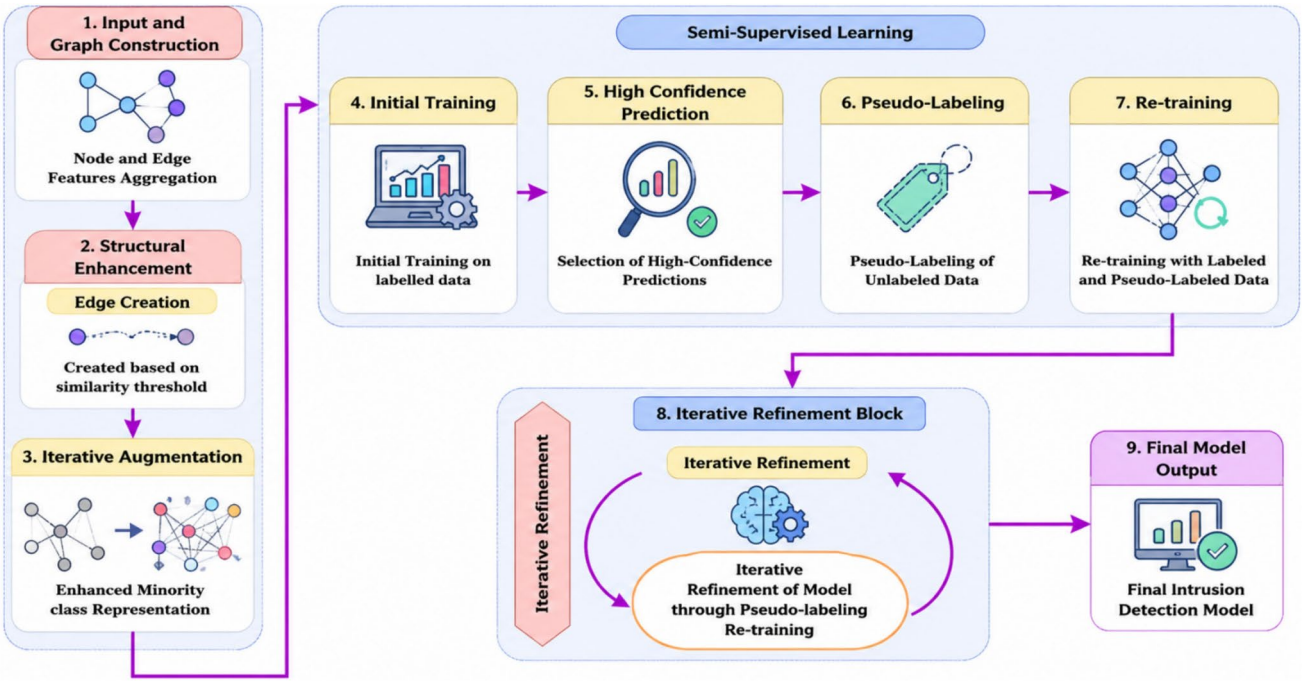


Fig. 1 SS-MRAGNet model's overall workflow

4.1.3 Multi-residual attention mechanism

The proposed approach's main innovation is the multi-residual attention mechanism, which is focused to aid the highly associated network's parts during the aggregation of data [35]. By means of the attention mechanism of scaled dot-product, the $\alpha_{ij}^{(l)}$ attention coefficients are calculated where the vector concatenation is given as || and the weight vector which is understandable is denoted as $a^{(l)}$. This mechanism computes $\alpha_{ij}^{(l)}$ using the Eq. (7).

$$\alpha_{ij}^{(l)} = \frac{\exp\left(Leaky\mathrm{ReLU}\left(a^{(l)}\left[W^{(l)}h_i^{(l-1)} \| W^{(l)}h_j^{(l-1)} \| W_e^{(l)}e_{ij}\right]\right)\right)}{\sum_{k\in N(i)}\exp\left(Leaky\mathrm{ReLU}\left(a^{(l)}\left[W^{(l)}h_i^{(l-1)} \| W^{(l)}h_k^{(l-1)} \| W_e^{(l)}e_{ik}\right]\right)\right)} \tag{7}$$

here, the learnable weight vector is denoted as $a^{(l)}$ for computing attention scores at layer . The $\alpha_{ij}^{(l)}$ detect every j neighboring node's influence on the node 's aggregation process. Within this mechanism, the residual interactions guarantee that global and local information are captured effectively by the approach in the graph. The proposed minimizes the gradient vanishing challenge while training and allows the network to learn network's deeper representations by implementing residual connections. Figure 2 illustrates the process of node and edge aggregation using multi-residual attention.

The multi-residual attention extends this mechanism across multiple attention heads $n \in N$ to capture both local and global structural dependencies.

$$H^{(l+1)} = \|_{n=1}^{N} \sum_{j\notin M(i)} \alpha_{ij}^{n,(l)} W_n^{(l)} h_i^{(l)} \tag{8}$$

In the above equation, $H^{(l+1)}$ refers to the output node embedding matrix at layer $l+1$, obtained after aggregating features from all attention heads. $\|_{n=1}^{N}$ is the concatenation operator that joins the outputs from all M attention heads to form the final node representation for the next layer. $M(i)$ refers to the set of neighbors directly connected to node in the IoT graph, $\alpha_{ij}^{n,(l)}$ is the normalized attention coefficient evaluated by the nth attention head at layer , and $W_n^{(l)}$ refers to the learnable weight matrix of the nth attention head at layer , used to alter the input features $h_i^{(l)}$ into a new feature space.

4.2 Self-supervised pre-training

This stage of pre-training was incorporated to improve the minority-class intrusion presentation. Auxiliary tasks are utilized in this stage for model pre-training over the IoT network structure before the model's fine-tuning for the task of categorization. The strong representations of features are learned by this pre-training approach, especially for classes that are underrepresented.

4.2.1 Edge augmentation

Edge augmentation includes producing extra edges among the nodes in the minority class for graphical representation improvement [36]. This helps capture the minority nodes' relationships, and the process is explained as below:

a) Identify Minority Node:

The identification of nodes that constitute to minority class is the initial step. This is performed through preliminary classification or clustering methods, which brings the underrepresented nodes' preliminary function.

b) Similarity Calculation:

Depending on the specific characteristics and the node features of IoT network, the similarity is calculated using different measures like Euclidean distance, and cosine similarity, as shown in Eq. (9).

$$Sim\left(v_i, v_j\right) = \frac{h_i.h_j}{\|h_i\|\|h_j\|} \tag{9}$$

where, the v_j and v_i nodes' cosine similarity is denoted as $Sim\left(v_i, v_j\right)$ and added e_{ij} new edge when this similarity overcomes τ, which a predefined threshold. The v_i and v_j nodes' feature vectors are represented as h_i

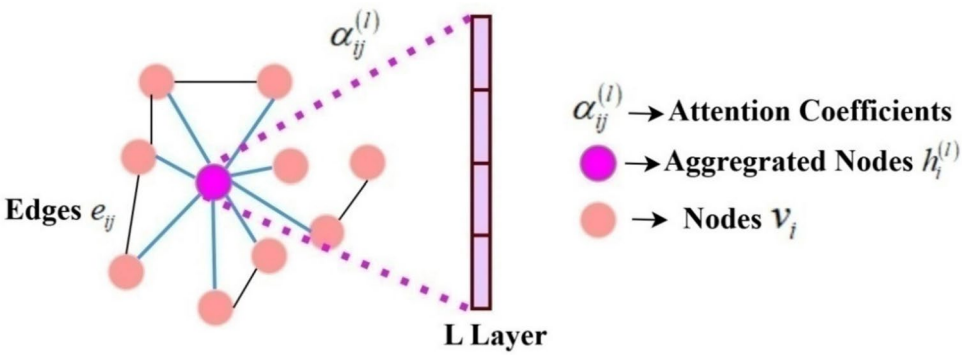


Fig. 2 Node and Edge Aggregation based on Multi-Residual Attention

and h_j respectively. The $||h_i||$ and $||h_j||$ are the Euclidean (L2) norms of node feature vectors.

c) Threshold Selection:

The threshold is chosen to detect the connection on which the nodes pair should be performed after the calculation of similarities. For the creation of an edge, above-threshold similarity score nodes are considered. The hyperparameter, which is the threshold, is tuned on the basis of the intrusion detection activity's particular requirements.

d) Edge Creation:

New edges are created between minority-class nodes after the identification of minority nodes. Euclidean distance or cosine similarity creates these edges by measuring similarity, guarantees new nodes-edges connection with same characteristics.

e) Iterative Augmentation:

In every augmentation round, iterative edge augmentation produces extra edges on the basis of features of an updated node, which guarantees that the augmented graph enhances its potential for minority class representation gradually.

f) Augmented Graph:

Then, the real graph G augmentation with the freshly formed edges produces a graph $G' = (V, E')$ called an augmented graph, where E' contains the edges of the augmented and original. For the proposed model's pre-training, the G' is utilized to study extra strong minority-class nodes. The edge augmentation for representation of enhanced minority-class is represented in Fig. 3.

4.2.2 Pseudo-labeling

Here, labels are assigned to the unlabeled data during the phase of pre-training [37]. Then, these pseudo-labels are utilized to enhance the classification ability. The initial predictions are produced by the model for assigning labels. Depending on the predictions that are in high confidence, this pseudo-labels assignment is done by,

$$\hat{y}_i = \underset{c \in C}{\operatorname{argmax}} p\left(c | h_i^{(l)}\right) \tag{10}$$

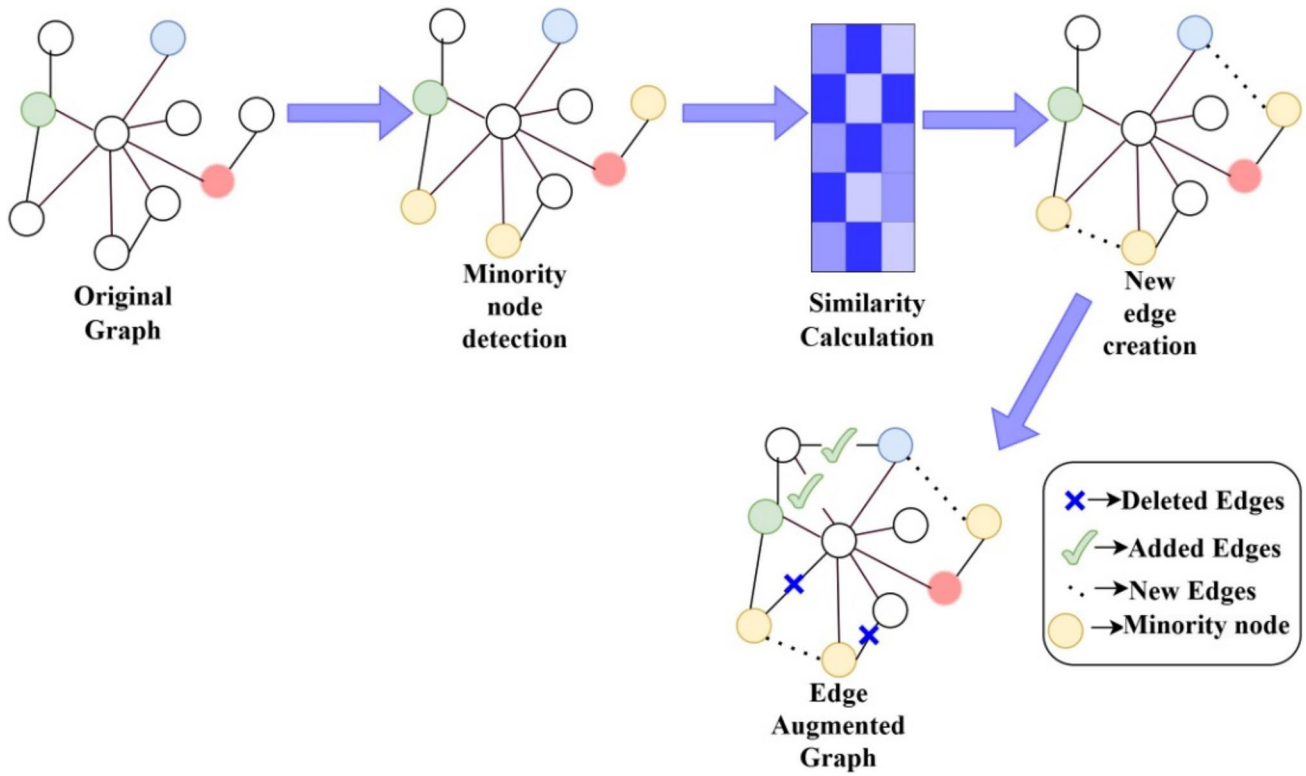


Fig. 3 Edge Augmentation for Representation of Enhanced Minority-Class

For nodes, the c class predicted probability is denoted as $p\left(c|h_i^{(l)}\right)$, and the assignment of pseudo-label to nodes is denoted as $\hat{y}_i$. The pseudo-labeling steps include:

a) Training of Initial Model:

The initial training on the labeled data aids to generate unlabeled nodes' forecasts.

b) Assignment of Pseudo-Label:

Using the forecasts, pseudo-labels are assigned to the unlabeled data by the high-confidence prediction nodes.

c) Model Re-Training:

On the integrated datasets, the model is re-trained, which contains the data that are pseudo-labeled and the original data that are labeled. This aids prediction refinement, specifically for classes in minority.

4.2.3 Self-supervised loss function

The loss of binary cross-entropy obtained for prediction of edge (BCE) and mean square error (MSE) obtained for clustering node is reduced by this pre-training phase.

$$L_{self-supervised} = \sum_{(i,j)\in E'} BCE\left(\hat{y}_{ij}, y_{ij}\right) + \lambda \sum_{v_i \in V} MSE\left(\hat{c}_i, c_i\right) \tag{11}$$

here, the edge existence's true label is denoted as y_{ij}, the edge existence's predicted probability is denoted as $\hat{y}_{ij}$, the assignment of a true cluster for node and the assignment of a predicted cluster are c_i and $\hat{c}_i$ respectively. Figure 4 illustrates the stage of self-supervised pre-training, including the process of edge augmentation and pseudo-labeling.

4.3 Classification through self-training

Self-training is termed as a process of iteration that refines predictions through multiple training rounds that enhances the accuracy of the approach [38]. The self-training steps are given as:

a) Training of Initial Model:

On the labeled data, the model is trained to produce both unlabeled and labeled nodes predictions, which is represented as:

$$\hat{y}_i = softmax\left(W_c h_i^{(l)} + b_c\right) \tag{12}$$

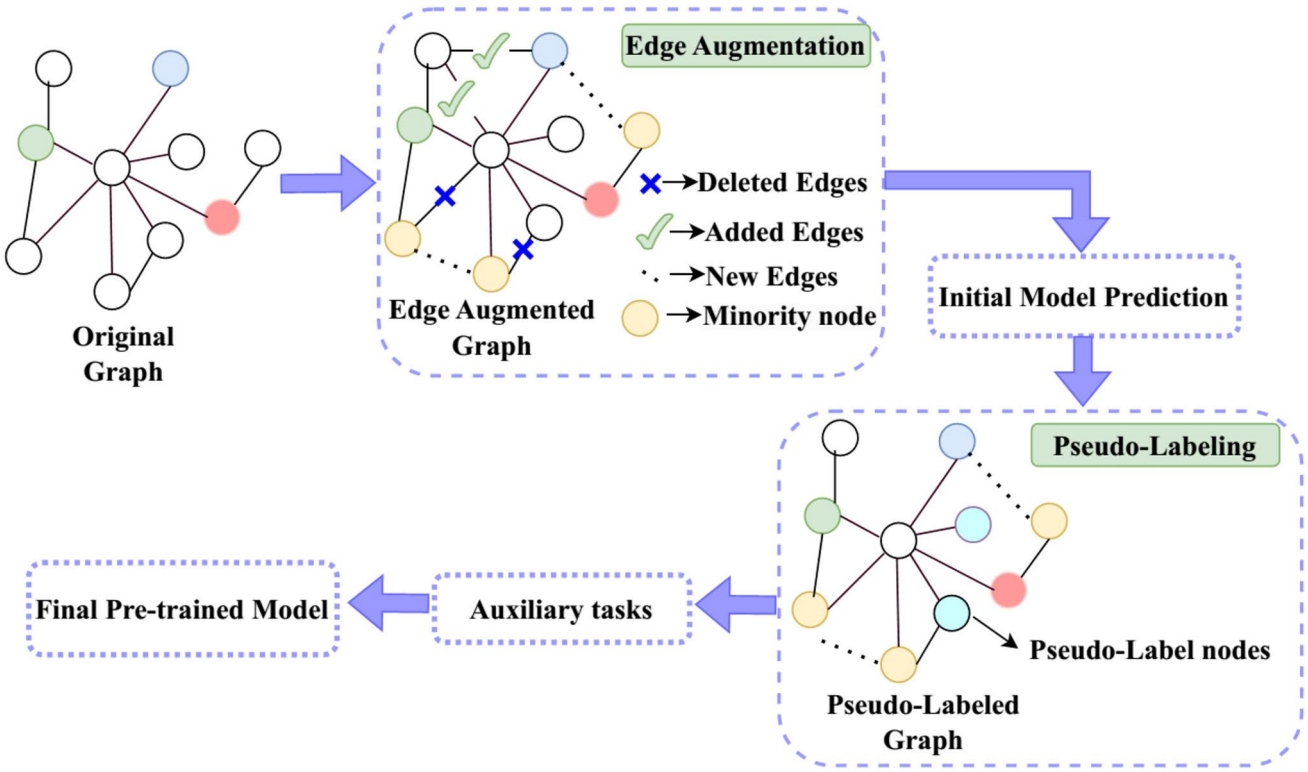


Fig. 4 Stage of Self-Supervised Pre-Training

here, the matrix of bias and weight for the layer of classification are represented as b_c and W_c respectively, and the probability distribution which is predicted on classes for node is $\hat{y}_i$.

b) Selection of High-Confidence:

The predictions of the model are estimated to detect high confidence prediction. The predictions in high confidence are referred to the assignments of high probability to a specific class. (eg: above a 0.9 threshold). The selection of predictions that are in high-confidence on the basis of τ_c confidence threshold:

$$\hat{y}_i is chosen if \max(\hat{y}_i) > \tau_c \tag{13}$$

For node, the maximum probability of prediction is represented as $\max(\hat{y}_i)$.

c) Model Re-Training and Pseudo-Labeling:

The allocation of pseudo-labels to the unlabeled data is performed by predictions that are in high confidence. Through the combination of the pseudo-labeled and labeled dataset, the re-training of the model is performed.

$$L_{classification} = -\sum_{i \in V} \sum_{c \in C} y_i^c \log \hat{y}_i^c \tag{14}$$

here, the pseudo label for c class and node is y_i^c, the c 's predicted probability is $\hat{y}_i^c$.

d) Iterative Refinement:

Here, the predictions of the model are refined in every round of iterative self-training, and the pseudo-labelled training data that are newly assigned are updated. Until the stabilization of the performance of the model, there is a continuous iterative refinement happens, denoting that the accuracy does not improve notably for further round of iterations. Figure 5 illustrates the process self-training iteration for the detection of minority class intrusion.

The sequential operations of the SS-MRAGNet framework are illustrated in algorithm 1.

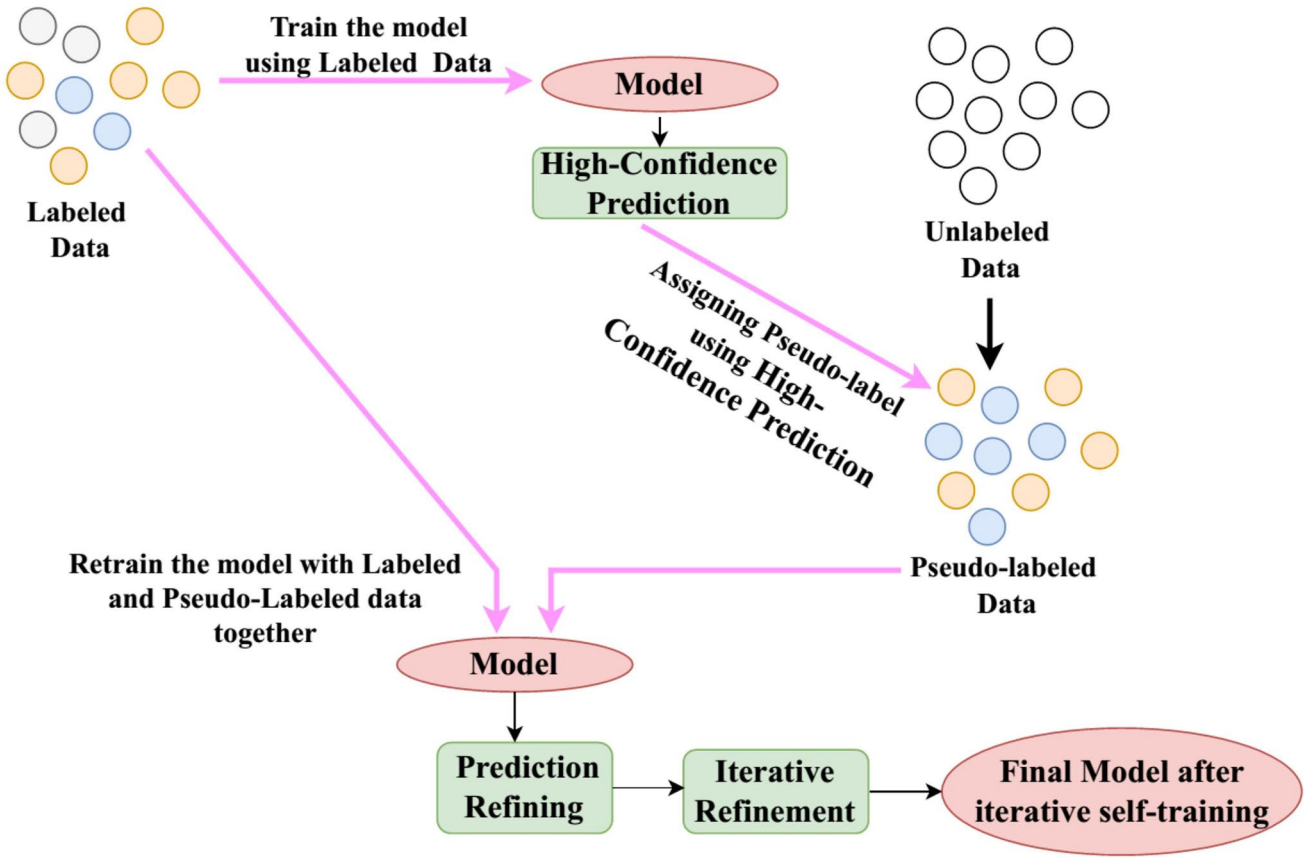


Fig. 5 Self-Training Iteration for the Detection of Minority-Class Intrusion

Algorithm 1: SS-MRAGNet Intrusion Detection Process

Input: $G=(V,E)$ denotes the Graph, Node features X. Edge features E_f, Confidence threshold θ, Number of iterations T.

Initialization: Randomly initialize model parameters Θ.

Step 1: Node and Edge Aggregation:

- For each node $v_i \in V$, aggregate neighboring node features and edge features using multi- residual attention.
- Compute attention weights using scaled dot-product attention.

Step 2: Self-Supervised Pre-training:

- Perform pseudo-labeling for unlabeled nodes with high-confidence predictions.
- Execute edge augmentation by connecting minority-class nodes based on cosine similarity.
- Update graph G' to include augmented edges.

Step 3: Self-Training:

- Train SS-MRAGNet using both labeled and pseudo-labeled data.
- Select high-confidence predictions $(p>\theta)$ and refine pseudo-labels.
- Iterate over T rounds until convergence.

Step 4 : Final Prediction:

- Use the trained model for classification on test data.

Output: Intrusion labels for all nodes in the graph.

The time and space complexities of the SS-MRAGNet components are described as follows. The complexity of node and edge aggregation is expressed as $O(Nd^2+Ed)$, where d represents the feature dimension, E represents the number of edges, and N represents the number of nodes. For multi-residual attention, the expression for complexity is $O(HNd^2)$, with attention heads H. The complexity of edge augmentation is $O(Nd^2)$, due to pairwise cosine similarity. The complexity for pseudo-labeling with the number of classes c is expressed as $O(NC)$. The complexity for training via gradient descent and total memory footprint are given as $O(TEd)$, per epoch for T epochs and $O(Nd+Ed+Hd^2)$ respectively.

5 Experimental results

The experiments presented below are structured to answer the research questions (RQ1–RQ4) introduced in Section 1, covering framework design, learning strategy, robustness, and deployment feasibility. The experimental design not only evaluates performance but also validates the necessity of the SS-MRAGNet for enhancing IoT security. Traditional IoT intrusion detection systems often fail to (i) maintain high recall for minority attack classes, (ii) remain robust under the condition of noisy and dynamic IoT traffic, and (iii) operate efficiently on resource-limited edge devices. The experiments presented here directly address

these unresolved issues. The evaluation metrics definition, short baseline description, and a highly comprehensive discussion of the findings are presented in the supplementary file (Part B).

5.1 Experimental setup

The proposed method is implemented in operating system called Microsoft Windows 10 with Intel(R) Core(TM) CPU i7-8750H 64-bit, and 32 GB RAM. Python 3.7.11 is the programming environment utilized by the proposed method. For the implementation of the deep learning framework, the TensorFlow 2.4.1 library is used. The Scikit-learn library is utilized for performing the classification process. Considering these platform specifications and system requirements, this research guarantees the reliable and efficient implementation for enhanced detection of minority-class intrusion.

To confirm rigorous evaluation and prevent information leakage, all preprocessing operations were carried out in a training-only and strictly ordered manner. At first, the splitting of dataset was occurred with the ratio in the order of 80% training and 20% testing subsets. The test set was isolated completely and was not utilized to do any preprocessing or model selection steps. Then, five-fold cross-validation was exclusively applied on the training subset of data. Within each fold of cross-validation, Min–Max normalization parameters were solely learned from the fold-specific training data and later implemented to the respective validation data. Random Under-Sampling was also carried out only on the fold-specific training data to rectify class imbalance issue, while validation data stayed untouched. After the selection of model, the training of final model was happen on the entire training set through undergoing same procedure, and the learned parameters of normalization were implemented to the set of held-out test for final evaluation. This pipeline setup guarantees a leakage-free and fair experimental protocol.

5.2 Parameter settings

The parameters that are set for the estimation of the SS-MRAGNet model's performance are listed in Table 1.

Table 1 Parameter settings

Parameters	Values
Loss Function	BCE
Batch size	32,768
Learning Rate	$[10^{-3},10^{-2}]$
Number of Attention heads	4
Optimizer	Adam
Epoch	100

All the parameters set up in the above table ensure the effectiveness of the proposed model for enhanced detection of minority-class intrusion.

5.3 Datasets description

Using four different datasets, such as CSE-CIC-IDS2018, ToN-IoT, CIC-Darknet2020, and UNSW-NB15, the SS-MRAGNet model is evaluated. The data from the datasets is split into two subgroups such as testing (20%) and training (80%). The details of the datasets are provided as follows.

- **CSE-CIC-IDS2018 dataset:** This dataset was created with the aim of intrusion detection evaluation, developed by the Canadian Institute for Cybersecurity (CIC) and a collaborative project between the Communications Security Establishment (CSE). Seven types of attack scenarios, such as infiltration, Botnet, DDoS, Brute-force, Web attacks, Heartbleed, and DoS, are included in this dataset. 80 flow based features are extracted from the dataset, which was collected through https://www.kaggle.com/datasets/solarmainframe/ids-intrusion-csv.
- **Ton-IoT dataset:** At the University of New South Wales (UNSW), this dataset is gathered from the Canberra Cyber of Australian Defense Force Academy (ADFA). Ton-IoT dataset consist of several features of network traffic utilized to estimate the effectiveness of different artificial intelligence dependent cybersecurity applications. From heterogeneous sources of IoT data, several cyberattacks such as DDoS, XSS, password cracking attacks, MITM, DoS, reconnaissance, injection attacks, backdoors, ransomware, and normal events are collected. Out of a total of 22,339,021 records, 461,043 records are utilized for training and testing process. https://www.kaggle.com/datasets/amaniabourida/Ton-IoT
- **CIC-Darknet2020 dataset:** CIC-Darknet2020 dataset is an academic network traffic categorization dataset. It is generated through the amalgamation of ISCXTor2016 and ISCXVPN2016 and links the corresponding Tor traffic and Virtual Private Network (VPN) with respect to Darknet categorization. The two classes of this datasets are normal and darknet traffic (https://www.kaggle.com/datasets/dhoogla/cicdarknet2020).
- **UNSW_NB15 dataset:** From the Australian Center for Cyber Security (ACCS)' Cyber Range Lab, UNSW_NB15 dataset is obtained in the year 2015. The Perfect-Storm tool is utilized for producing traffic that is benign. The dataset consist of 9 kinds of attacks, like reconnaissance, fuzzes, generic, exploits, backdoors, worms, DoS, analysis, shellcode, and normal. 49 extracted characteristics of network traffic comprises of the network packet and host investigation on the basis of the flow

between them to differentiate abnormal and normal behaviors. Out the total of 2,540,044 records, selected 257,673 records for the testing and training process. The data was gathered through https://www.kaggle.com/datasets/mrwellsdavid/UNSW_NB15.

5.4 Data preprocessing

The proposed framework utilized three preprocessing steps like data cleaning, class balancing, and data normalization.

- **Data Cleaning:** Within the datasets, the process of identifying and addressing inaccuracies and errors to confirm their reliability and accuracy is known as data cleaning. In order to maintain data quality and reliability, this process is very important. Different activities like inconsistent entries correction, missing values management, dealing with outliers, and duplicates elimination are included in data cleaning. Methods like prediction, imputation, and deletion are utilized to rectify missing values. To neglect redundancy, duplicates are identified and removed. Uniformity is ensured by standardizing the differences in formatting or spelling, which is known as inconsistent entries.
- **Class Balancing:** Class Balancing is performed by utilizing the technique called Random Under-Sampling (RUS). In this technique, some dominant class samples are eliminated randomly until it attains with minority class alignment for balancing classes. It neglects poor results caused due to the noise by neglecting samples from the class of majority. This technique is applicable during domination of majority class over the minority class. It is significant to note that RUS does not constitute the primary imbalance handling mechanism and is only used as a fold-specific training stabilization step. The primary imbalance handling mechanism is addressed through the proposed self-supervised and graph-based learning strategies.
- **Data normalization:** Data normalization is essential to remove inconsistent range of value that causes bias in particular algorithms. This method aids to narrows values' ranges and prepares the collected data for effective computation. Min–Max Normalization is utilized for the scaling of values from 0 to 1. The expression is given below.

$$C_{new} = \frac{C - \min(C)}{\max(C) - \min(C)} \tag{15}$$

In the above equation, the new scaled value is denoted as C_{new}, the original value is denoted as C, the dataset's minimum and maximum value is denoted as $\min(C)$ and $\max(C)$ respectively.

5.5 Evaluation metrics

The SS-MRAGNet framework is evaluated using different evaluation metrics are accuracy, precision, recall, F1-Score, and Area Under the Receiver Operating Characteristic (AUC-ROC) curve [39].

5.6 Performance analysis

The performance analysis facilitates understanding the effectiveness of the proposed model. Figure 6 depicts the four different datasets' confusion matrix. It is in the form of a table formulated between the values of predicted and actual. Figure 6 (a) represents the confusion matrix for the actual and predicted values of CSE-CIC-IDS2018 dataset's

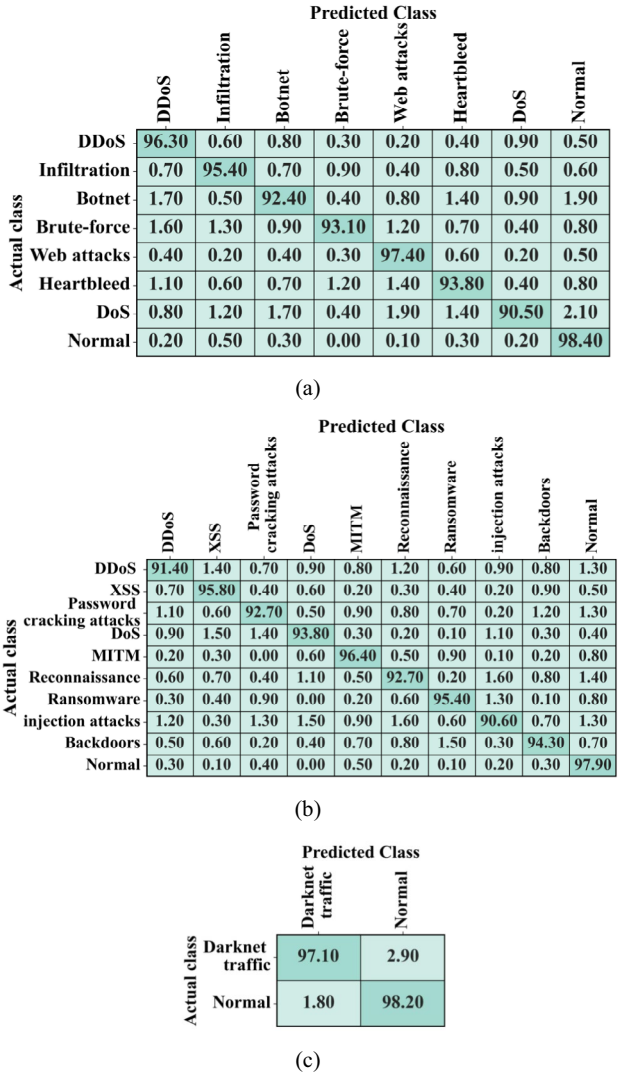


Fig. 6 (**a**) CSE-CIC-IDS2018 Confusion Matrix (**b**) Ton-IoT Confusion Matrix (**c**) CIC-Darknet2020 Confusion Matrix (**d**) UNSW_NB15 Confusion Matrix

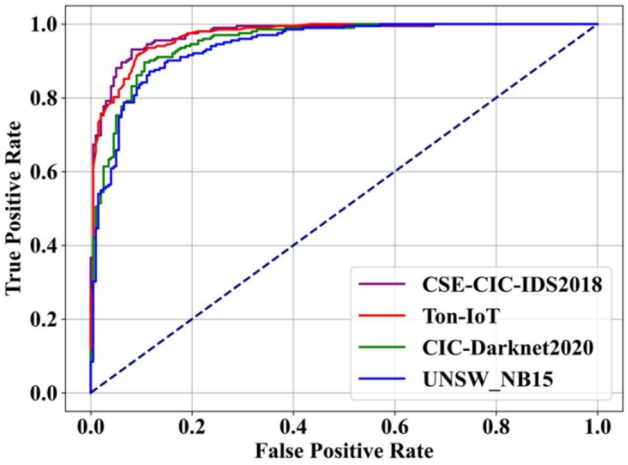


Fig. 7 AUC-ROC analysis of different datasets

8 classes, Fig. 6 (b) represents the confusion matrix for the actual and predicted values of Ton-IoT dataset's 10 classes, Fig. 6 (c) represents the confusion matrix for the actual and predicted values of CIC-Darknet2020 dataset's 2 classes, and confusion matrix is represented in the Fig. 6 (d) for the actual and predicted values of UNSW_NB15 dataset's 10 classes.

Figure 7 depicts the graphical representation of the AUC-ROC curve of different datasets. The AUC-ROC analysis is performed depending on false and true positive values. Higher AUC-ROC performance is obtained from the four different datasets for the proposed model. The gained values of AUC-ROC are provided as 0.988, 0.983, 0.987, and 0.985 for CSE-CIC-IDS2018, UNSW_NB15, Ton-IoT, and CIC-Darknet2020 datasets respectively.

The analysis of training and testing accuracy and loss is represented in Fig. 8. Figure 8 (a) illustrates the training and testing accuracy analysis with respect to epochs. During the process of training, the obtained value of accuracy at 100 epochs is 0.98. During the process of testing, the values of accuracy of 0.72, 0.8, 0.84, 0.87, and 0.88 are obtained from various epochs. Figure 8 (b) illustrates the analysis of training and testing loss in terms of epochs. From the graph, the proposed SS-MRAGNet model's testing loss is greater than that of the loss obtained during training. The losses during testing and training were obtained as 0.12 and 0.02 respectively at 100 epochs.

5.7 Robustness evaluation

This experiment addresses RQ3 by analyzing model stability under noise and class-imbalance perturbations. It examines robustness under label noise, Gaussian perturbations, and imbalance to demonstrate stable performance in realistic IoT conditions. The robustness is tested against different noise types with respect to various metrics are depicted in Fig. 9. Figure 9 (a) illustrates the robustness analysis under various noises with respect to accuracy, precision, F1-Score, and recall. Figure 9 (b) illustrates the robustness analysis under various noises with respect to AUC-ROC. The kind of random noise added into the nodes' characteristics for the simulation of malfunctions is referred as Gaussian noise. The label noise flips a particular percentage of the labels of the training set that stimulates mislabeled data. For Dropout noise, random edges and nodes from the graph are removed for simulating offline devices or missing connections. The outcomes denote that the proposed approach is sensitive to dropout noise since the removal of edges and nodes minimizes its capability of capturing essential relationships.

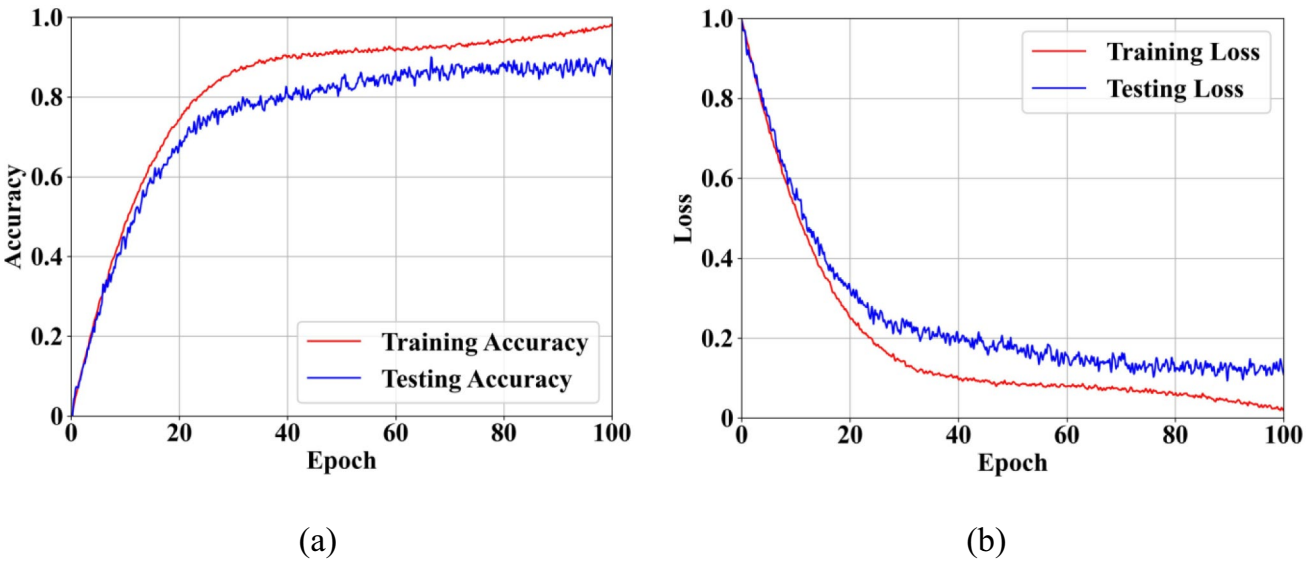


Fig. 8 Analysis of training and testing with respect to (**a**) accuracy (**b**) loss

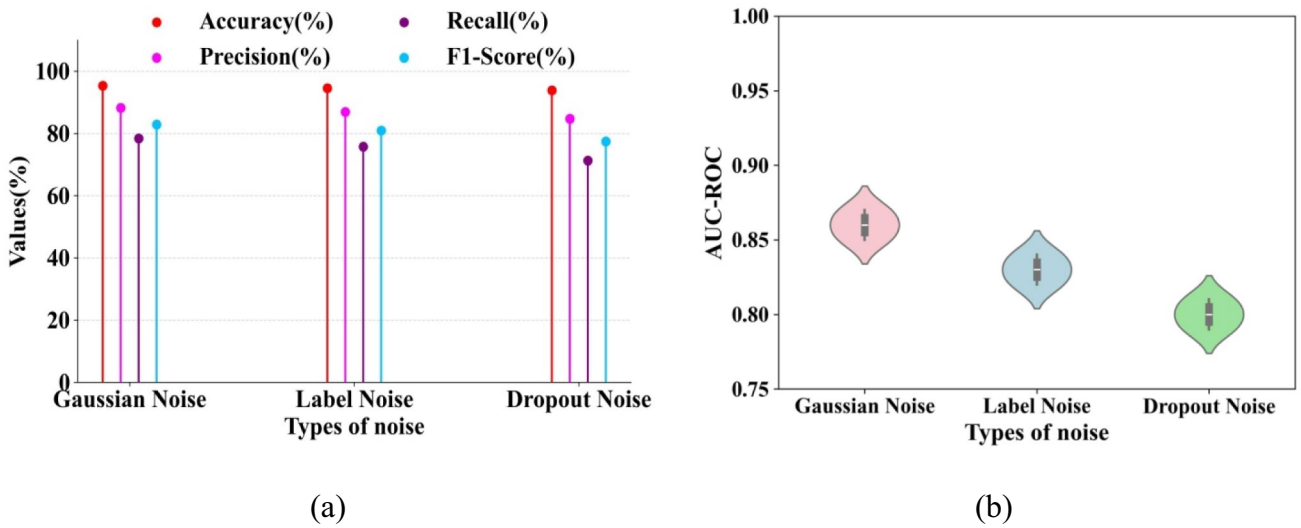


Fig. 9 Robustness analysis under various noises on (**a**) Accuracy, Precision, F1-Score, Recall, (**b**) AUC-ROC

Under three scenarios, the level of class imbalance is tested. In the analysis of first level, the class consists of 90% benign and 10% malicious and the next level of class imbalance is considered with 5% malicious and 95% benign in the second scenario. For effective analysis, 2% of malicious and 98% benign are considered in the third scenario. This robustness analysis using various imbalance scenarios on all performance measures is depicted in Fig. 10. Figure 10a illustrates the robustness analysis under varying imbalance scenarios with respect to accuracy, precision, recall, and F1-Score. Figure 10b illustrates the robustness analysis under varying imbalance scenarios with respect to AUC-ROC. From the figure, the performance of SS-MRAGNet decreases gradually as there is an increase in class imbalance, but still, its performance is better under extreme levels of imbalance. Recall decreases under the scenarios of higher class imbalance, but its higher values scoring capability for the F1-score determines the proposed model's robustness.

In order to know about the robustness of the proposed method, a five-fold cross-validation test is conducted. This test provides a highly reliable estimation of the proposed method's effectiveness on unseen data. It aids in ensuring the ability of generalization and minimizing the data variability impact of the SS-MRAGNet model. Table 2 shows the cross-validation results of four datasets for every fold. The mean values of the datasets were computed to reduce the data partitioning variance.

This part examines RQ1 by evaluating the effect of removing core components such as multi-residual attention and edge augmentation. In order to check the effectiveness of the proposed model, a sequence of ablation studies is performed to determine every component's contribution to the framework of SS-MRAGNet. Particularly, it investigates the components such as the node and edge aggregation, the multi-residual attention mechanism, edge augmentation, and self-supervised pre-training. The detailed ablation study is presented in Table 3. Model A showed better scores on all metrics, which is the SS-MRAGNet model. Model B signifies the study using node and edge aggregation without including the multi-residual attention mechanism. Model A has the capability to handle both node and edge data in comparison with model B; therefore multi-residual attention mechanism is highly effective for the proposed approach to aggregate the data. Then, model C is the study conducted without the inclusion of edge augmentation among the minority-class's nodes. When compared with model C, model A attained better outputs, which indicates that the inclusion of edge augmentation is essential to capture minority nodes' inter-relationships for rare intrusion detection. At last, model D represents the study conducted without the stage of self-supervised pre-training, and it is represented as. The outcome shows that the model D's performance is lesser than model A for every performance metric. Hence, this stage is significant to pre-train the model to increase the representation of minority-class intrusion.

5.8 Baseline comparisons

The question RQ2 is evaluated by comparing the proposed SS-MRAGNet with conventional supervised baselines on multiple IoT intrusion datasets under class-imbalance conditions. It compares SS-MRAGNet with traditional GNN and CNN-based IDS models to show improvements in minority-attack detection and generalization. The methods utilized for comparison are IDSBPSO [14], EIDM [15], LS-PIO [16], IMOPSO-IDS [18], BT-TPF [19], GMC-WAE [25], FeCo [26], and TL TinyML-based IDS [27]. For the confirmation of a fair and transparent comparison,

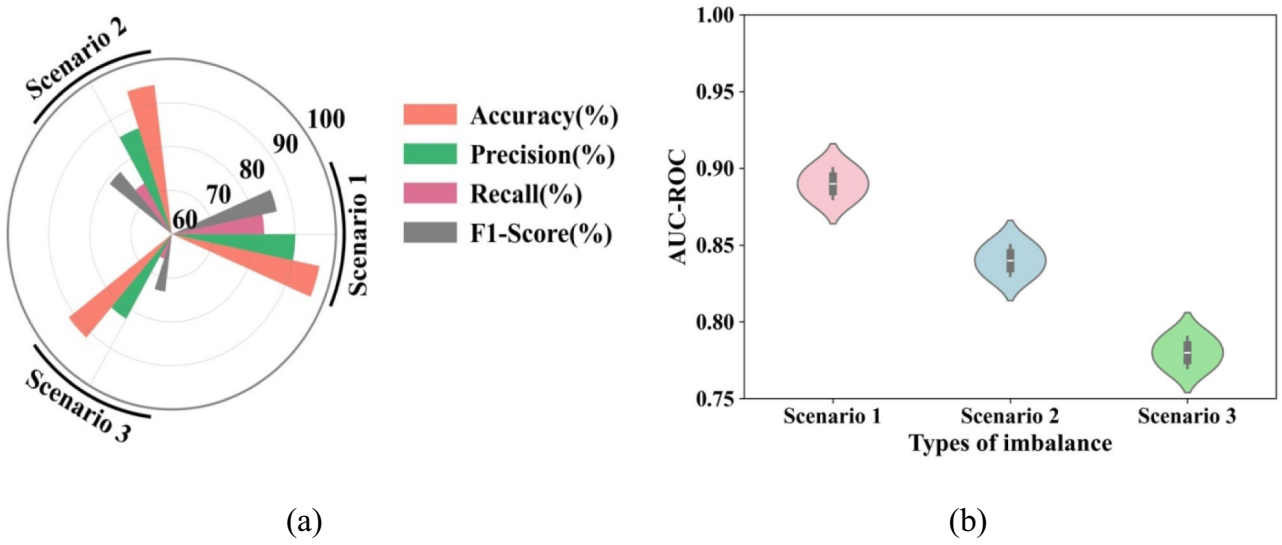


Fig. 10 Robustness analysis under varying imbalance scenarios on (**a**) Accuracy, Precision, Recall, F1-Score (**b**) AUC-ROC

Table 2 Five-fold cross-validation on training sets of datasets

Folds	CSE-CIC-IDS2018	Ton-IoT	CIC-Dark-net2020	UNSW_NB15
1	97.8	97.1	98.4	96.35
2	97.9	97.7	98.6	96.45
3	98.7	98.6	98.7	97.7
4	98.9	98.7	98.8	98.3
5	99.8	98.9	99.2	98.4
Average	98.62	98.2	98.74	97.44
Std deviation	0.816	0.687	0.296	0.987

all baseline results were obtained by re-implementing and evaluating using the same datasets, preprocessing pipeline (underwent identical data cleaning, normalization, and class balancing procedures to neglect preprocessing bias), same train–test split (80:20), and under the same experimental settings, and equivalent performance metrics as SS-MRAGNet. Further, SS-MRAGNet outperforms baseline methods across multiple datasets that indicates performance gains of the proposed architecture rather than experimental bias.

Table 4 shows accuracy, precision, recall, and F1-Score results obtained through four different datasets for existing and proposed models. The SS-MRAGNet model obtained accuracy of 98.25% denoting robust categorization performance compared to other models for the four different datasets. The introduced model improved the identification of intrusions in the minority-class compared to all other models. Through analyzing four datasets, the introduced model obtained a precision of 98.21%, outperforming other models. Also, a recall of 98% and 98.1% F1-score is obtained, which signifies the effectiveness of the SS-MRAGNet model in identifying intrusions in the minority-class.

Table 3 Ablation study

Models	Measures				
	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Model A	98.25	98.21	98	98.1	0.985
Model B	93.34	82.25	67.42	76.56	0.822
Model C	94.16	83.72	70.59	76.43	0.831
Model D	94.83	85.94	74.28	79.67	0.856

The graphical representation of the AUC-ROC curve for the existing and SS-MRAGNet model is depicted in Fig. 11. This curve is plotted between the true positive and false positive rates. The plots aided in determining the best classification threshold. The proposed SS-MRAGNet model achieved an AUC of 0.985 when compared with existing models.

The complexity analysis corresponds to RQ4, examining the computational feasibility of deploying SS-MRAGNet in the environment of IoT. This analysis estimates memory, runtime, and scalability to confirm its applicability for the deployment of edge-gateway. Table 5 illustrates the SS-MRAGNet method's computation complexity. The complexity analysis of runtime and memory determines that SS-MRAGNet attains efficiency in computation for the implementation of IoT. It maintains better accuracy by achieving the lowest memory consumption (0.87 GB) and fastest inference time (53 ms) among all compared models. The reduced FLOPs (40 G) and moderate parameter count (7.3 M) denote a lightweight model structure suitable for the environments of edge or fog computing. Compared with existing approaches, SS-MRAGNet requires less than half their memory footprint and one-fifth of their training time, confirming its feasibility for real-time or near-real-time IoT intrusion identification tasks.

Table 4 Results of accuracy, precision, recall, and F1-Score obtained through four different datasets for existing and proposed models

Methods	CSE-CIC-IDS				ToN-IoT				CIC-DarkNet				UNSW-NB15			
	Accuracy (%)	Precision (%)	Recall(%)	F1-Score (%)	Accuracy (%)	Precision (%)	Recall(%)	F1-Score (%)	Accuracy (%)	Precision (%)	Recall(%)	F1-Score (%)	Accuracy (%)	Precision (%)	Recall(%)	F1-Score (%)
IDSBPSO	95.72	95.89	96.17	96.02	89.93	93.84	93.21	93.52	91.83	89.93	94.69	92.24	92.53	88.24	94.86	91.43
EIDM	96.38	96.52	96.94	96.72	91.66	98.06	85.49	91.34	96.42	88.42	92.43	90.38	95.26	94.18	87.62	90.78
LS-PIO	93.01	93.96	94.23	94.09	95.39	97.29	93.51	95.36	97.09	94.27	88.97	91.54	96.82	87.95	85.94	86.93
IMOPSO-IDS	97.26	98.01	97.88	97.94	94.28	94.32	94.66	94.48	94.58	92.18	95.48	93.8	93.78	97.42	87.46	92.17
BT-TPF	94.59	95.12	95.99	95.55	96.74	95.67	95.58	95.62	92.11	90.99	92.91	91.93	94.81	96.46	93.27	94.83
GMCWAE	86.4	82.4	86.7	84.49	87.5	82.9	85.46	84.16	87.48	81.41	85.7	83.49	87.7	80.57	85.9	83.14
FeCo	89.5	93.6	95.1	94.34	90.9	92.8	94.1	93.44	87.6	91.6	93.7	92.63	90.6	92.8	95.1	93.93
TL TinyML-based IDS	89.2	93.6	95.1	94.34	90.4	92.8	92.1	92.44	88.2	91.9	93.3	92.59	90.5	92.8	93.1	92.94
SS-MRAGNet	98.69	98.68	98.65	98.66	98.23	98.43	97.25	97.83	97.88	97.56	98.41	97.98	98.21	98.17	97.69	97.92

Table 6 also displays the statistical analysis of various existing and proposed methods using the Friedman ranking test. From the table, it was observed that the performance of proposed method is better than the existing methods in terms of Accuracy, Precision, Recall, F1-score, and AUC-ROC, and also the SS-MRAGNet method attained top rank in comparison with all other existing methods. The Friedman ranking test yielded chi-squre value χ_{η}^{2} as 26.78 and its corresponding ρ_{η} as 0.0008, which is less than the significant threshold $\sigma_{\eta} = 0.05$. The degree of freedom is obtained as 8(9–1), and its corresponding critical value is 15.507, and the value of ρ_{η} is 0.05. The outcome is statistical significant since $26.78 > 15.507$.

6 Discussion

The results demonstrate that SS-MRAGNet fills critical gaps faced by existing IDSs. While traditional models lose detection sensitivity under imbalance and noise, SS-MRAGNet sustains high recall (≈98%), AUC-ROC even with only 5% minority-class samples and 20% injected noise. Its design utilizes self-supervised learning and mechanism of multi-residual attention, offering resilience and feature richness that are lacking in traditional GNNs and CNNs. These outcomes prove that the SS-MRAGNet is a necessary and reliable advancement for the deployment of IoT intrusion identification. Although the SS-MRAGNet framework is evaluated on general IoT intrusion datasets, its graph representation naturally supports distributed or device-to-device communication structures, making it adaptable to diverse IoT topologies.

6.1 Discussion and implications

The robustness evaluation demonstrates that a high score of AUC-ROC is maintained by the proposed approach even when the input data has 20% noise and maintained high recall, even when the minority class was made with only 5% training data, confirming its resilience to noisy and highly imbalanced IoT environments. As evidenced by ablation studies, these gains stem from the complementary contributions of the model components: multi-residual attention selectively emphasizes discriminative traffic patterns, self-supervised pre-training increases generalization under limited labels, and edge augmentation increases sensitivity by reinforcing structural consistency among minority-class nodes. Compared with conventional GNN and baseline models that handle nodes and edges uniformly, SS-MRAGNet utilizes attention-driven feature selection and self-learned representations to gain consistently superior effectiveness across datasets. From a practical perspective,

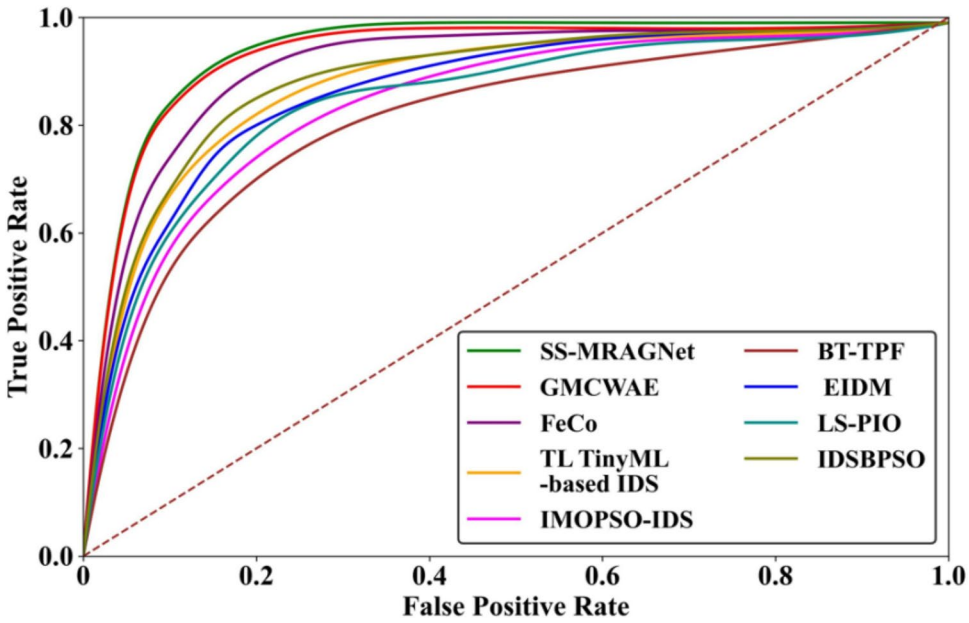


Fig. 11 Comparative analysis of the AUC-ROC curve with respect to existing and proposed methods

Table 5 Complexity analysis and statistical analysis using the Friedman ranking test of various methods

Methods	Parameter (M)	Flops (G)	Memory consumption (GB)	Inference time (ms)	Training time (mins)
IDSBPSO	9.6	63.12	2.95	98	258.4
EIDM	13.72	45.69	1.34	63	549.7
LS-PIO	8.59	93.78	1.28	74	101.7
IMOPSO-IDS	14.63	99.49	1.14	85	514.9
BT-TPF	17.39	112.46	2.5	57	364.7
GMCWAE	35.55	139.2	3.6	69	314.2
FeCo	47.29	91	1.22	65	246.1
TL TinyML-based IDS	30.79	85.89	1.46	72	205.2
SS-MRAGNet	7.3	40	0.87	53	96.9

the model's graph-dependent design stimulates effective utilization of network topology and device metadata, increasing coordinated attack detection while decreasing false positives. These properties make SS-MRAGNet well applicable for real-world applications like IoT, IIoT, smart city, and healthcare security. While deployment at scale introduces drawbacks associated with computational overhead, resource constraints, and evolving threat patterns, these can be rectified through lightweight model variants, distributed processing, and adaptive or continual learning extensions. Future improvements incorporates temporal and spatial context to further increase robustness against over dynamic and geographically distributed attacks for reinforcing the model's suitability for real-world IoT security deployment.

6.2 SS-mragnet's significance in minority-class detection

This subsection evaluates RQ2 by examining whether the integration of self-supervised learning and the mechanism of multi-residual attention improves the capability of the model to detect minority and rare intrusion types without relying on extensive labeled datasets. Through various combinations of attention and self-supervised strategies, these limitations are addressed by the SS-MRAGNet model:

- The detection ability that are rarely happening events is improved by means of multi-residual attention, which learns the highly related features of the edge and node, specifically for intrusions in the minority class.
- The model's robustness is enhanced by the self-supervised pre-training through learning minority nodes' representation of features, which are mostly less common in IoT datasets.
- Further, reinforcement of minority node representation is done by edge augmentation, permitting the model to add connections among the nodes, or else be separated in the graph structure.

6.3 Mechanistic interpretation of results

The results indicate that the SS-MRAGNet shows robust performance under various settings of perturbation. However, the impact differs depending on the introduction of the regularization type or the noise included. While the dropout mechanism reduces overfitting, severe most fall in

Table 6 Friedman ranking test results

Methods	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Mean Rank
IDSBPSO	92.5%	91.97%	94.73%	93.3%	0.91	5
	6	8	2	5	4	
EIDM	94.93%	94.29%	90.62%	92.3%	0.93	6
	4	4	8	7	7	
LS-PIO	95.57%	93.36%	90.66%	91.98%	0.92	6
	2	5	7	8	8	
IMOPSO-IDS	94.97%	95.48%	93.87%	94.59%	0.95	3.4
	3	2	5	2	5	
BT-TPF	94.56%	94.56%	94.43%	94.48%	0.94	4.2
	5	3	4	3	6	
GMCWAE	87.27%	81.82%	85.94%	83.82%	0.98	7.6
	9	9	9	9	2	
FeCo	89.65%	92.7%	94.5%	93.59%	0.97	4.8
	7	7	3	4	3	
TL TinyML-based IDS	89.57%	92.77%	93.4%	93.07%	0.96	6
	8	6	6	6	4	
SS-MRAGNet	98.25%	98.21%	98%	98.1%	0.985	1
	1	1	1	1	1	

recall during the application of graph-attentive layers. This is attributed to the connectivity loss within minority-class subgraphs, dropping feature connections disorders the critical signals flow between sparsely connected nodes. In the aggregation process of self-attention, such random omission of edges or vertices breaks attention pathways that are significant for minority-class feature propagation, causing the model to miss subtle patterns of anomaly and hence shows lowered recall. Conversely, label noise displayed a different pattern of degradation because of its interaction with the process of self-training. Since pseudo-labels are refined iteratively, initial label errors tend to be amplified through self-training, leading to error propagation and poor calibration of class boundaries. There is a chance of reinforcing incorrect labels, which minimizes generalization and explains the observed instability under higher noise ratios.

In the meantime, the addition of Gaussian feature noise caused only a modest performance drop. Unlike dropout or label noise, the graph's topological integrity is maintained by Gaussian perturbations; the edge connectivity and underlying neighborhood stay intact, permitting the attention mechanism to effectively reweigh the nodes that are in informative. The noise acts as a mild regularizer that increases strength instead of spoiling relationships in the structure. On the whole, these outcomes signify that the model's reliability depends on perturbations of input and components interactions of graph attention and self-supervised learning with noise. The maintenance of structural coherence and label calibration accuracy is essential for sustaining recall and entire quality of detection in IoT network circumstances.

6.4 Limitations

Though the SS-MRAGNet model has many advantages, it also faces some limitations such as:

- **Scalability:** The computation of the multi-residual attention mechanism is costly for larger graphs. This could bring threats to the detection of real-time intrusion, as the scale of IoT networks develops. Large-scale implementations may require optimizations to handle vast amounts of data efficiently.
- **Complexity of Hyperparameter Tuning and Computational Requirements**: The SS-MRAGNet model requires careful hyperparameter tuning to obtain optimal results. The complexity of tuning needs extra resources for computation and skill while deploying the model. Additionally, the model includes multi-residual attention and self-supervised training stages that may impose high memory, processing power, and high computational overhead, particularly when deployed on resource-constrained IoT devices.
- **Generalization across Different Network Topologies:** The SS-MRAGNet model's performance over various network topologies requires further testing, as it was examined on different IoT datasets. Additional modifications or tuning are required for the model to be adopted in the environments of IoT, which are highly varied. Self-supervised learning highly relies on the available data's distribution and quality. If certain patterns of attack are underrepresented in the data of training, achieving generalization becomes hard by the model.

- **Real-Time Performance**: In IoT environments, intrusion identification necessitates real-time processing to decrease threats effectively. Conversely, only on public datasets, the model was evaluated. Its effectiveness in live systems of IoT with real-time traffic and noise remains unexamined. The SS-MRAGNet method may experience latency problems in scenarios of high-traffic, necessitating further improvements to increase inference speed.
- **Adversarial Robustness Unassessed**: The model's resistance to evasion or poisoning types of adversarial attacks was not determined and stays as a challenge for crucial applications. This critical gap, especially for applications that are security-sensitive where attackers deliberately may handle data to bypass identification.

6.5 Future research directions

The future research is listed below:

- **Scalability Improvements**: Exploring optimizations to the multi-residual attention mechanism is the future work could increase the scalability of SS-MRAGNet for IoT larger networks. Techniques like graph sampling or sparse attention could be determined to reduce overhead in computation.
- **Real-Time Detection**: While the performance of SS-MRAGNet is good on static datasets, further work is required to adapt the model in the environment of dynamic IoT for real-time identification of intrusion. This may in turn incorporates streaming data and develops approaches to manage continuous updating to the network of graph.
- **Transfer Learning**: The benefits of transfer learning are examined to implement proposed approach to different IoT networks could results applicability enhancement over diverse circumstances. This could involve pre-training of model on large and generic IoT datasets before fine-tuning it on particular activities or network topologies.
- **Adversarial Robustness**: Developing techniques for adversarial training specific to GNN could increase the model's security in circumstances where attackers may attempt to manipulate the structure of graph or features of node. Also, estimating the model's strength under adversarial conditions like attacks of evasion and poisoning remains a high priority for increasing acceptability in high-risk circumstances.

The process of SS-MRAGNet iterations will explore extensions in spatio-temporal, which incorporates embeddings of spatial and temporal to identify complicated intrusions including the patterns of location-specific timing in future. This will permit tracking behavior in real-time over time-ordered graphs, simulating predictive mechanism for security in dynamic circumstances. Also, future work will also explore the utilization of automated hyperparameter optimization to decrease overhead during tuning and increase efficiency of deployment.

7 Conclusion

The crucial role of NIDS in IoT network is to examine malicious behavior like security attacks. In this sense, this work presented SS-MRAGNet model for enhanced identification of network intrusion, and it is designed using an advanced GNN that effectively integrates aggregation of node and edge with a mechanism of multi-residual attention. The integration of edge augmentation, pseudo-labeling, and pretraining enhances balanced intrusion detection performance. To carry out final classification through refinement of prediction, the stage of self-supervised pre-training is followed by a process of self-training. The implementation of edge augmentation represents the minority-class nodes' additional edges in the graph. The model was estimated on four different intrusion detection datasets and displayed improved accuracy (98.25%), recall (98%), precision (98.21%), and F1-score (98.1%) compared to other existing methods. Unlike previous frameworks of IDS that struggle under the condition of real-world imbalance and noise, SS-MRAGNet determines high-fidelity, consistent detection of intrusion over various datasets of IoT. This necessitates a practical and strong remedy for next-generation IoT security. The model maintained strong performance under different kinds of noise and levels of class imbalance, guaranteeing its robustness and ability of generalization. Also, ablation studies ensured the necessity of every element in the SS-MRAGNet architecture, and statistical analysis aided the consistency of the model's effectiveness.

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Declarations

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